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Batch: A-2
Div: A

Subject: Programming with Java

ASSIGNMENT No : 2

TITLE : Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management. Methods and Static Members.

Problem Statement

Develop a Java application demonstrating the use of default, parameterized, and copy constructors for object initialization and data management.

Learning Objectives

To understand and implement default, parameterized, and copy constructors for object initialization.

Theory

Constructors are special member functions that are automatically invoked when an object is created. Java supports default constructors provided by the compiler, parameterized constructors for initializing objects with user-defined values, and copy constructors (implemented manually) for creating duplicate objects. Constructors improve code readability and simplify object initialization while supporting object-oriented programming principles.

Output Screenshots

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp2/Main.java
Defining class in java :
name : Devansh, age : 18

Learning static and final variable difference :
Object : 1
Object : 2
Object : 3
Counter : 3

Learning Instance Variable :
Name : Devansh -- ID : 10

Calling static function in java :
Sum : 20

Inner and Outer classes :
Inner Class Method
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Exercise

Q1. Create a Student class using default and parameterized constructors to initialize student name and roll number.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp2/Exercise/Exp2.java
Student Details :
Name : Devansh
Age  : 18
ID   : 101
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q2. Develop a Mobile Phone Inventory System using different constructors to initialize mobile details and create duplicate object records.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp2/Exercise/Ex2.java
Mobile Details :
Brand   : Samsung
CPU     : Qualcomm Snapdragon 8 Elite 2
RAM     : 12 GB
Storage : 256 GB
Price   : Rs.145000

Mobile Details :
Brand   : Samsung
CPU     : Qualcomm Snapdragon 8 Elite 2
RAM     : 12 GB
Storage : 256 GB
Price   : Rs.145000

dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Conclusion

This assignment helped in understanding the concept and importance of constructors in Java. It demonstrated the use of default, parameterized, and copy constructors for object initialization and data management. The implementation improved my understanding of object-oriented programming by showing how constructors simplify object creation, initialize data efficiently, and enhance code readability. These concepts are fundamental for developing well-structured, reusable, and maintainable Java applications and are widely used in real-world software development.